

Prob 3.: Two loads are connected in parallel across a voltage source of $100\angle 0^\circ V_{rms}$, where load 1 absorbs $500VAR$ at 0.6 lagging p.f. while load 2 absorbs $400VA$ at 0.75 leading p.f

1. The impedance of load 1 (Z_1), in Ω , is:
2. The current of load 2 (I_2), in Amps, is:
3. The complex power (S) supplied by the source is:
4. The power factor of the equivalent load is:
5. The capacitance needed to raise the pf to unity is:
6. The current supplied by the source, after the power factor is corrected, is:

(12 marks)

Prob. 4: Consider the circuit shown in Fig. 3,

1. The equivalent impedance seen by the source, Z_{in} , is:
2. The current supplied by the source, I_1 , is:
3. The load current, I_2 , is:
4. The voltage across the primary winding, V_1 , is:
5. The average power absorbed by the 2Ω resistor is:

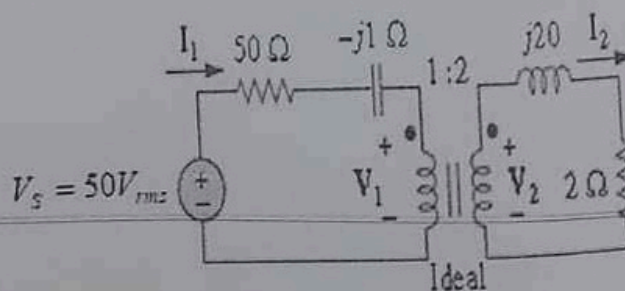


Fig.3

(10 marks)

- Prob. 1: Consider the circuit shown in Fig. 1,
1. The load impedance (Z_L), in Ω , that absorbs the maximum average power is:
 2. The open circuit voltage between a - b terminals (V_o), when the load impedance is disconnected, is:
 3. The short circuit current (I_{sc}), when a - b terminals are shorted is:
 4. The maximum average power transferred to the load impedance (Z_L) is:

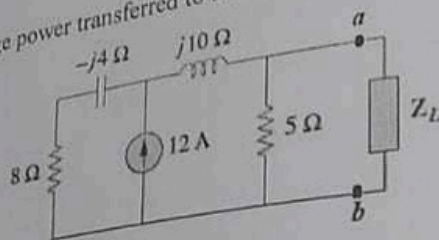


Fig.1

(8 marks)

Prob. 2: Consider the circuit shown in Fig. 2 and using mesh analysis method,

1. Write the two mesh equations, do not simplify:
2. The current I_1 is:
3. The current I_2 is:
4. The capacitor current, I_C , is:

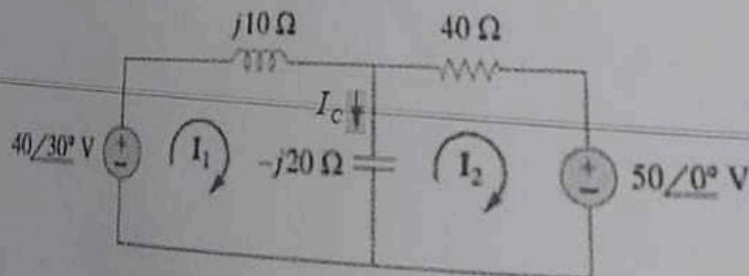


Fig 2

(8 marks)